

ATEF ABDELRAHMAN SHABAN HASSAN, MD

Physician | Clinical Researcher | Medical AI Developer | Founder, ResBot.ai | COO, OrthoGlobe Initiative

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Egyptian national | Date of birth 26 March 1999 | Based in Nasr City, Cairo

RESEARCH IMPACT AT A GLANCE

- **Peer-reviewed journal articles:** 33 (10 first-authored, 3 senior-authored), 2024 – 2026
- **Conference abstracts, proceedings and posters:** 9 (AHA, ASN Kidney Week, ESMO Immuno-Oncology, AAN, ISAKOS)
- **Preprints and protocol registrations:** 3
- **Verified peer reviews completed:** 9 across 7 international journals (Springer Nature SNAPP)
- **Citation metrics:** 71 citations, h-index 5 (Google Scholar, May 2026) — current figures at ORCID / Google Scholar
- **Medical AI products:** ResBot.ai (founder), CiteSort.AI, Oracle-RCC and 7+ further imaging / prediction models; 3 systems in development
- **Leadership:** Founder & Lead Engineer, ResBot.ai; Chief Operating Officer, OrthoGlobe Initiative (London, UK)

PROFESSIONAL PROFILE

Licensed medical doctor and prolific clinical researcher with broad experience across clinical and translational research spanning cardiovascular medicine, neurology, urology, infectious diseases, anesthesiology, orthopaedics, radiology and rheumatology. Author or co-author on 33 peer-reviewed journal publications — including systematic reviews, network meta-analyses, individual patient data and diagnostic test accuracy meta-analyses, randomized controlled trials, target trial emulations, and machine-learning studies — plus 9 conference abstracts and proceedings. Strong skills in reproducible data analysis (Python/R/Stata), study design, evidence synthesis, and extensive hands-on experience analysing large real-world clinical databases including the US National Inpatient Sample (NIS/HCUP), TriNetX Global Collaborative Network, and the UK Biobank for observational, comparative-effectiveness, and target-trial-emulation studies. Specialisation in applied medical AI — causal machine learning, fairness-aware clinical prediction, medical imaging deep learning, and LLM/NLP research automation. Founder and Lead Engineer of ResBot.ai, an end-to-end AI systematic review and meta-analysis platform, Chief Operating Officer of the OrthoGlobe Initiative (London), and an active peer reviewer for Springer Nature journals.

RESEARCH HIGHLIGHTS

- **Medical AI first/senior-author work** — causal ML for personalised surgical decision-making in aortic dissection with fairness auditing (Annals of Vascular Surgery, 2026); calibrated, explainable ML on routine laboratory data across 12,085 rheumatology patients (BMC Rheumatology, 2025); AI in surgical training (BMJ Leader, 2026); AI in bone, joint and soft-tissue infection management (Current Infectious Disease Reports, 2026; The Lancet Microbe, 2025).
- **Advanced causal and real-world evidence methods** — population-based target trial emulation in de novo metastatic prostate cancer (World Journal of Urology, 2026) and a nationwide comparative-effectiveness study of urinary decompression timing in 9,172 septic patients (BMC Urology, 2026).

- **Full range of evidence-synthesis designs** — pairwise, network, individual patient data (IPD), and diagnostic test accuracy meta-analyses, published across BJU International, Catheterization and Cardiovascular Interventions, Microsurgery, Abdominal Radiology, Neuroepidemiology, and Thrombosis Journal.
- **Clinical trial and prospective work** — co-author of a randomized trial of mini-PCNL versus RIRS for lower-calyceal stones (African Journal of Urology, 2025) and a pilot feasibility study of autologous fascia lata slings for stress urinary incontinence (LUTS, 2026).
- **Platform builder** — architect and lead engineer of ResBot.ai, a multi-model AI platform automating the entire systematic review pipeline from PICO search strategy to publication-ready manuscript, with a deterministic statistical engine and full PRISMA 2020 coverage.
- **International collaboration** — co-authorship with OrthoGlobe (London), Al-Azhar University, the Medical Research Group of Egypt, Harvard-affiliated methodologists, and ISAKOS/AAOS-fellowship surgeons.

EDUCATION

MB BCh, Bachelor of Medicine and Surgery

Faculty of Medicine (Cairo), Al-Azhar University, Cairo, Egypt | Enrolled 2017 – graduated December 2023 session; result ratified by University Council 18 April 2024

- Overall grade: Very Good with Honours — 5,272 / 6,300 marks (83.6%)
- Faculty nationally accredited by the National Authority for Quality Assurance and Accreditation of Education (NAQAAE)
- Clinical rotations: Internal Medicine, Surgery, Pediatrics, Obstetrics/Gynecology, Diagnostic Radiology, Emergency Medicine, Family Medicine
- Degree certificate serial HH0171494 (issued 2 July 2024); attested by the Egyptian Ministry of Higher Education

CLINICAL EXPERIENCE

Rotating Internship Physician (House Officer)

Al-Hussein University Hospital, Al-Azhar University, Cairo, Egypt | 1 March 2024 – 28 February 2025 (12 months, completed in full)

- Two months each: Internal Medicine, General Surgery, Pediatrics, Obstetrics & Gynecology, and Emergency Medicine & Critical Care
- One month each: Neurology and Neurosurgery
- Patient assessment, documentation, and participation in multidisciplinary care under supervision
- Daily correlation of clinical presentation with plain radiography, ultrasound, CT and MRI; requesting and interpreting imaging under supervision and presenting findings on ward rounds
- Certificate of completion no. 288 issued by the Department of Physician Affairs, Al-Azhar University Hospitals, 28 February 2025

General Practitioner

Al-Hussein Hospital, Cairo, Egypt | January 2023 – January 2024

- Provided comprehensive primary care services to 30+ patients daily in the outpatient department
- Performed physical examinations, ordered and interpreted diagnostic tests and first-line imaging, and developed treatment plans
- Managed chronic disease patients including diabetes, hypertension, and cardiovascular conditions; onward referral of complex cases
- Collaborated with multidisciplinary teams; maintained accurate electronic medical records and ensured timely follow-up

Medical Intern — Karmoz Medical Insurance Hospital, Alexandria, Egypt | Oct – Dec 2022

- Internal Medicine and Urology departments; ECG interpretation and hands-on basic ultrasonography

Medical Intern — Damanhur General Hospital, Egypt | Jul – Sep 2022

- Emergency Medicine rotation; triage, trauma radiography interpretation, suturing, IV access, resuscitation and stabilisation of critical patients

Medical Intern — Dumyat (Damietta) General Hospital, Egypt | Apr – Jun 2022

- Surgery, Pediatrics and Internal Medicine (emergency room and endoscopy unit); peri-operative care, well-child visits, basic surgical procedures

LICENSURE, REGISTRATION & VERIFICATION

- **Egyptian Ministry of Health practice license #367599** — valid 26 March 2025 to 26 March 2030.
- **Egyptian Medical Syndicate registration #363353** — Behira branch, registered 26 March 2025.
- **Dataflow Primary Source Verification** — complete, overall result POSITIVE with no derogatory records (case S003-2504-2401814, issued 10 May 2025); MBBCh degree and MOH license both verified. Issued to SCFHS; available for sharing with DOH Abu Dhabi, DHA and QCHP.
- **Immunisation, malpractice cover and health screening** — available to complete to employer specification prior to a start date.

RESEARCH, LEADERSHIP & SERVICE

- **Founder & Lead Engineer, ResBot.ai (2025 – present)** — Conceived and built an end-to-end AI-powered systematic review and meta-analysis platform. Own the full architecture: an 8-stage AI pipeline, a 4-tier multi-model routing stack, a deterministic statistical engine, and the web application. Lead product, technology and clinical-validation strategy. Co-founded with Prof. Mohamed A. Imam (Professor of Digital Health, University of East London) and Dr. Ahmed Negida (postdoctoral research fellow, Harvard Medical School).
- **Chief Operating Officer (COO), OrthoGlobe Initiative (London, UK)** — Lead day-to-day operations and research delivery for an international orthopaedic research network; coordinate multidisciplinary teams across the UK, Ireland and other European centres; drive AI-driven musculoskeletal research initiatives; oversee documentation, timelines, and quality control.
- **Founder, CiteSort.AI** — AI-assisted citation screening and PRISMA-aligned study triage platform; GPT-powered screening with blind dual-review, duplicate detection, Cohen's kappa, and PubMed/Scopus integration (predecessor project to ResBot.ai).
- **Medical Researcher, Al-Azhar University (2021 – present)** — Design research protocols and statistical analysis plans; lead and contribute to systematic reviews, meta-analyses and machine-learning studies; collaborate with international multi-centre research teams.
- **Collaborator, Medical Research Group of Egypt (MRGE) / Negida Academy (2021 – present)** — Cross-institutional systematic review and meta-analysis collaborations.
- **Peer reviewer, Springer Nature journals (SNAPP)** — 9 verified reviews completed across 7 journals (see Peer Review Experience).

PEER REVIEW EXPERIENCE

Active peer reviewer for international biomedical journals via the Springer Nature SNAPP platform. 9 verified reviews completed across 7 journals:

- **Journal of Robotic Surgery (Springer Nature)** — 3 reviews
- **BMC Cardiovascular Disorders** — 1 review

- **BMC Urology** — 1 review
- **BMC Geriatrics** — 1 review
- **BMC Endocrine Disorders** — 1 review
- **BMC Nephrology** — 1 review
- **World Journal of Urology (Springer Nature)** — 1 review

Selected reviews completed

- **BMC Cardiovascular Disorders** — Purine xanthine oxidase inhibitors and the prognosis of chronic heart failure: a meta-analysis. **Recommendation: Reject.** Identified critical methodological flaws, including inappropriate pooling of RCTs and observational studies, missing heterogeneity statistics, and overstated causal claims.
- **Journal of Robotic Surgery** — Safety and Outcome Assessment of Robot-Assisted Artificial Urinary Sphincter Implantation in Stress Urinary Incontinence: Systematic Review and Single-Arm Meta-Analysis. **Recommendation: Major revision.** Provided a detailed re-analysis plan including stratification by sex/site/indication, Hartung–Knapp random-effects modelling, prediction intervals, and GRADE assessment.
- **BMC Urology** — Risk factors for postoperative delirium in elderly patients undergoing urologic surgery: systematic review and meta-analysis. **Recommendation: Major revision.** Flagged PRISMA inconsistencies, non-standard quality-based exclusion, effect-measure mismatches, and provided a remediation plan with leave-one-out and prediction intervals.
- **BMC Geriatrics** — Frailty and postoperative complications in elderly patients undergoing cardiac surgery: systematic review and meta-analysis. **Recommendation: Revise.** Identified internal inconsistencies in continuous-outcome effect sizes (ICU stay, LOS) and a non-standard MACCE definition.
- **BMC Endocrine Disorders** — Glucagon-like peptide-1 receptor agonists in heart failure: an umbrella meta-analysis. **Recommendation: Major revision.** Flagged absence of safety outcomes despite stated scope, transparency gaps in the least-redundant set construction, and overstated conclusions given low-to-moderate GRADE certainty.
- **Journal of Robotic Surgery** — Robotic-Assisted Ureteric Reimplantation Versus Laparoscopic and Open Approaches: Systematic Review and Meta-Analysis. **Recommendation: Major revision.** Identified procedural-scope contamination (pyeloplasty/ureteroureterostomy studies pooled with ureteric reimplantation), retrospective protocol registration, and potential overlap with prior work.
- **BMC Nephrology** — Prevalence of Accessory Renal Arteries in Africa — Cadaveric and Imaging Studies: Systematic Review and Meta-Analysis. **Recommendation: Major revision.** Identified a critical statistical error (variance reported in place of standard error, underestimating SEs ~25-fold) and recommended Freeman–Tukey double-arcsine transformation with full re-analysis.
- **World Journal of Urology** — Effect of Music on Pain during Cystoscopy: Systematic Review and Meta-Analysis. **Recommendation: Major revision.** Flagged an unfilled GRADE placeholder, missing publication-bias assessment, undiscussed MCID, and conceptually heterogeneous inclusion (binaural beats classified as 'music').
- **Journal of Robotic Surgery** — Comparing Feasibility and Safety of Robotic-Assisted vs Manual Carotid Artery Stenting: Systematic Review and Meta-Analysis. **Recommendation: Reject.** Identified critical data contamination from a mixed-procedure source study (~60% of pooled patients did not undergo CAS), pervasive numerical discrepancies, and contradictions between stated and executed statistical methods.

CORE SKILLS

Evidence Synthesis & Biostatistics

- Protocol development and registration (PROSPERO/OSF), PRISMA 2020 reporting, screening, data extraction, risk of bias, GRADE certainty assessment
- Pairwise, network, diagnostic test accuracy, individual patient data (IPD), and umbrella meta-analyses

- Risk-of-bias instruments: ROB 2, ROBINS-I, ROBINS-E, QUADAS-2, QUADAS-C, PROBAST+AI, NOS, JBI, AMSTAR-2
- R (meta, metafor, survival), Stata, Review Manager (RevMan); random-effects models, Hartung–Knapp correction, prediction intervals, leave-one-out and influence diagnostics, publication-bias testing, reproducible analytical workflows

Real-World Data & Large Clinical Databases

- US National Inpatient Sample (NIS) — HCUP: cohort construction, ICD-10 code extraction, weighted survey analysis, outcomes and cost analytics
- TriNetX Global Collaborative Network: propensity-score matched cohort studies, time-to-event analyses, pharmacoepidemiology, and safety-signal detection across multi-institutional EHR data
- UK Biobank: phenotype definition, genetic and imaging data linkage, and large-scale association and prediction modelling
- Target trial emulation, propensity-score methods, Cox proportional hazards and competing-risks modelling
- Designed and executed multiple clinical observational studies using these databases, from protocol to publication-ready analyses

Programming & Data Science

- Python: pandas, NumPy, matplotlib, Seaborn, scikit-learn, TensorFlow/Keras, PyTorch, OpenCV, scikit-image
- R: statistical analysis and visualisation (meta, metafor, survival); Stata
- Web and app development: Flask, FastAPI, Streamlit, REST APIs, Vue.js, Tailwind CSS, Bootstrap; interactive dashboards with Plotly and Dash
- SQL, Git, reproducible computational environments; cloud tooling on AWS (S3, EC2) and Google Cloud AI; IDEs: VS Code, PyCharm, Jupyter

Machine Learning & Medical AI

- Supervised ML, model evaluation, probability calibration, explainability (SHAP), fairness auditing across demographic subgroups, model cards
- Causal machine learning and heterogeneous treatment effect estimation for clinical decision support
- Deep learning for medical imaging (CNN-based classification and segmentation; EfficientNet, U-Net; transfer and self-supervised learning); Transformer-based architectures
- Survival prediction modelling and multi-institutional external validation
- NLP and LLM-based research automation; retrieval-augmented generation (RAG), multi-model routing, structured extraction, hallucination control via chain review

Clinical

- Comprehensive patient assessment, diagnosis, and treatment planning
- Basic surgical procedures, wound management, suturing, IV insertion
- ECG interpretation, laboratory result analysis, basic imaging interpretation and bedside ultrasonography
- Emergency medicine: triage, basic life support, trauma management, critical care
- Chronic disease management: diabetes, hypertension, asthma, COPD
- Electronic medical records and hospital information systems

Research Operations

- Manuscript preparation, reference management, journal submission and peer-response handling
- Cross-institutional collaboration and stakeholder communication; Microsoft Office and Adobe Acrobat

DIAGNOSTIC IMAGING & DIAGNOSTIC-ACCURACY EXPERTISE

Diagnostic Accuracy Methodology

- Diagnostic test accuracy (DTA) meta-analysis: bivariate and hierarchical summary ROC models, pooled sensitivity and specificity, index-test versus reference-standard design
- QUADAS-2 and QUADAS-C risk-of-bias appraisal for imaging and diagnostic studies
- Published DTA meta-analysis of standalone 29-MHz micro-ultrasound for clinically significant prostate cancer (Abdominal Radiology, 2026 — senior author)
- Interpretation and reporting of diagnostic performance metrics: likelihood ratios, predictive values, calibration, and net clinical benefit

Imaging Artificial Intelligence

- Deep learning for medical image classification and segmentation; model evaluation for imaging: ROC/AUC, calibration, decision-curve analysis, SHAP explainability, and fairness auditing
- Published work on AI in musculoskeletal and infection imaging (The Lancet Microbe 2025; Current Infectious Disease Reports 2026) and AI in surgical training (BMJ Leader 2026)
- AI governance in the reading room: automation bias, PROBAST+AI appraisal, and STARD / CLAIM / TRIPOD+AI reporting standards

Clinical Imaging Skills

- Systematic interpretation of chest, abdominal and skeletal plain radiographs; recognition of acute surgical, trauma and pediatric findings
- Basic ultrasonography including focused abdominal and obstetric scanning
- Requesting CT and MRI with clinical justification and correlating cross-sectional findings under supervision; radiation protection, ALARA and appropriateness criteria

MEDICAL AI & RESEARCH AUTOMATION PROJECTS

ResBot.ai — Founder & Lead Engineer (resbot.ai)

End-to-end AI-powered systematic review and meta-analysis platform — from search strategy to publication-ready manuscript in one workspace, with human oversight at every step.

- **8-stage AI pipeline** — AI search strategy (PICO-based Boolean query generation, MeSH auto-suggestion, multi-database search across PubMed, Scopus, OpenAlex, Semantic Scholar, CrossRef and ClinicalTrials.gov) → smart title/abstract screening → full-text pipeline → data extraction studio → risk of bias → meta-analysis → GRADE and synthesis → manuscript generation.
- **4-tier multi-model AI architecture** — task-specific model routing: a fast tier for high-throughput screening, a long-context tier for full-paper extraction, and an independent reasoning tier from a different vendor family for chain review, which independently verifies every decision to catch fabrication and accuracy errors before they reach the manuscript.
- **Deterministic statistical engine** — 11 effect measures (OR, RR, RD, MD, SMD, HR, proportions, correlations, DOR), 7 between-study variance estimators (DerSimonian–Laird, REML, Paule–Mandel and others), Hartung–Knapp correction, Egger regression for publication bias, 13 interactive chart types, and R code generation for meta/metafor — real mathematics, not model-generated p-values.
- **Risk of bias & GRADE** — 9 validated RoB instruments plus a custom-tool builder, with AI-assisted domain assessment where every judgment links back to the highlighted evidence passage in the source PDF; five-domain GRADE certainty rating and an auto-generated, Cochrane-compatible Summary of Findings table.
- **Full-text acquisition & extraction** — automatic PDF retrieval from 15+ academic sources (Unpaywall, PMC, CrossRef, Semantic Scholar, OpenAlex, Europe PMC, bioRxiv, CORE, DOAJ and others), AI OCR recovery for

scanned documents, adaptive schema discovery, and 24 supported extraction data types across text, tables and figures.

- **Reporting & governance** — full PRISMA 2020 coverage (27/27 items), complete provenance trail with confidence scores on every extraction, blind multi-reviewer screening with inter-rater reliability, RAG-enhanced assistant with voice interaction, and DOCX/LaTeX/XLSX export.
- **Co-founders** — Prof. Mohamed A. Imam (Consultant Trauma & Upper-Limb Surgeon; Professor of Digital Health, University of East London) as Clinical Director and Dr. Ahmed Negida (postdoctoral research fellow, Harvard Medical School) as Methodology Lead.

Other AI Projects

- **Oracle-RCC** — multi-institutional AI survival prediction model for renal cell carcinoma, developed and externally validated across 19,579 patients (SSRN preprint, 2025; ESMO Immuno-Oncology proceedings 2024/2025).
- **RCC Overall-Survival Predictor (web app)** — Flask/Bootstrap application serving Random Forest and Gradient Boosting survival models for renal cell carcinoma, with Plotly/Dash performance dashboards.
- **Knee arthroplasty decision tool** — machine-learning model predicting progression to knee arthroplasty in osteoarthritis; ISAKOS Congress 2025 ePoster and IJAIA 2025 paper.
- **CiteSort.AI** — AI-assisted citation screening and study triage for systematic reviews; GPT-powered screening, smart duplicate detection, blind dual screening, Cohen's kappa, PubMed/Scopus integration; PRISMA-aligned structured export.
- **MedResearch AI Lab** — RAG-based research assistant with voice interaction and eight specialised tools for screening, extraction templates, and manuscript utilities.
- **IntelliSpine** — deep-learning pipeline for vertebral segmentation and pathology detection on spinal imaging (TensorFlow, CNN-based segmentation).
- **OsteoFuture** — calibrated, explainable ML for osteoporosis, fracture, and nonunion risk stratification in orthopaedic decision support.
- **Dental panoramic lesion classifier** — EfficientNet-B0 deep learning model for classification of intrabony lesions on panoramic radiographs (Research Square preprint, 2025).
- **OCT Retinal Classifier** — PyTorch/TensorFlow CNN classifying OCT scans into CNV, DME, drusen and normal for diabetic retinopathy and macular degeneration detection; Vue.js/Tailwind front end.
- **Medical AI Toolkit** — open-source collection of pre-trained models and utilities for medical image analysis.

In Development

- **AI-Powered Clinical Decision Support System** — integrated platform leveraging multi-modal ML for diagnostic and management recommendations.
- **Surgical Outcome Predictor** — machine-learning model for preoperative surgical outcome and complication prediction.
- **Living systematic review engine and Medical Literature Synthesizer** — continuous-update reviews and advanced NLP synthesis built on the ResBot.ai pipeline.

EARLY RESEARCH TRAINING (2022 – 2023)

Systematic reviews and meta-analyses initiated as a medical student and intern under the Medical Research Group of Egypt; several progressed to the peer-reviewed publications listed below.

- Cardiology / cardiothoracic surgery: N-acetylcysteine for postoperative atrial fibrillation (published Minerva Cardiology and Angiology, 2025); network meta-analysis of CTEPH treatments (published AHJ Plus, 2024; Atherosclerosis abstract 2024); dapagliflozin in heart failure with reduced ejection fraction; eplerenone vs spironolactone (OSF registration 2024, published BMC Cardiovascular Disorders, 2024)

- Neurology: risk of dementia after early-diagnosed depression (150+ studies screened); revascularisation for moyamoya disease (updated meta-analysis); PDE-5 inhibitors and Alzheimer's disease (published Neurological Sciences, 2024)

PEER-REVIEWED PUBLICATIONS

Authored / co-authored papers in reverse chronological order. Author name shown in bold. Total: 33 peer-reviewed journal articles. Verified list: ORCID 0000-0003-3393-9105.

2026

1. Cook P, **Hassan A**, Kerstein R, Imam MA. *Embracing artificial intelligence in surgical training: a clinical leadership perspective*. **BMJ Leader**. 2026; online first (leader-2025-001258). DOI: 10.1136/leader-2025-001258 | PMID: 42431830
2. Dogha MM, **Hassan AA**, Mohamed OM, Farag M, Nouh IS, et al. *Early urinary decompression and mortality in septic obstructing ureteral stones: a nationwide comparative-effectiveness study*. **BMC Urology**. 2026; 26(1):155. DOI: 10.1186/s12894-026-02119-z | PMID: 42399868 | National Inpatient Sample (n = 9,172)
3. Salman MFA, **Hassan AA**, Wasfy IS, Deif H, Hindawy MA, et al. *Autologous Transobturator Fascia Lata Sling for Female Stress Urinary Incontinence: A Pilot Feasibility and Safety Study*. **LUTS: Lower Urinary Tract Symptoms**. 2026; 18(5):e70085. DOI: 10.1111/luts.70085 | PMID: 42521462
4. Hassaan AM, **Hassan AA**, Abdelkhalik WS, Saad AM, Hosny AS, et al.; Imam MA. *Efficacy and safety of MIPO versus ORIF in distal tibial fractures: a systematic review and meta-analysis*. **Journal of Orthopaedics and Traumatology**. 2026; 27(1):30. DOI: 10.1186/s10195-026-00938-2 | PMID: 42303945
5. **Hassan AA**, Aboelsaad AY, Abdel Gawad AM, El-Dydamony EM, Algammal MI, et al. *Early vs. delayed primary local therapy and survival in de novo metastatic prostate cancer: target trial emulation*. **World Journal of Urology**. 2026; 44(1):409. DOI: 10.1007/s00345-026-06489-9 | PMID: 42240667 | First author; population-based target trial emulation
6. Lotfy AA, **Hassan AA**, Goda AS, Abd El Baky DL, et al. *Intravenous Lidocaine as an Adjunct to Propofol Sedation for Endoscopic Retrograde Cholangiopancreatography: A Systematic Review and Meta-Analysis*. **Current Anesthesiology Reports**. 2026; 16(1):4. DOI: 10.1007/s40140-026-00707-1
7. Bakry HE, Abdellah MI, Khalaf HA, Assem AA, ElGamal MM, Abdo MI, Elkasaby MH, **Hassan AA**. *Efficacy and safety of direct oral anticoagulants in pediatric congenital and acquired heart disease: a systematic review and meta-analysis of randomized controlled trials*. **Thrombosis Journal**. 2026; 24(1):43. DOI: 10.1186/s12959-026-00850-z | PMID: 42032668 | Senior author
8. Abouelmagd ME, et al.; **Hassan AA**. *Efficacy and safety of apomorphine in the treatment of Parkinson's disease: a systematic review and meta-analysis of randomized controlled trials*. **Therapeutic Advances in Neurological Disorders**. 2026; 19:17562864261436541. DOI: 10.1177/17562864261436541
9. **Hassan AA**, Elshami MI, Shaban Soliman AA, Abdo Ibrahim MI, Hassan Ali MH, et al. *Causal Machine Learning for Personalized Surgical Decision-Making in Aortic Dissection: A National Fairness-Aware Clinical AI Framework*. **Annals of Vascular Surgery**. 2026; 129:1–16. DOI: 10.1016/j.avsg.2026.03.031 | PMID: 41903841 | First author; causal ML + fairness auditing on national data
10. Youssof HA, **Hassan AA**, Saleh AA, Shafiea A, Mohammed AM, et al. *Preemptive versus preventive intravenous acetaminophen/ibuprofen fixed-dose combination after robot-assisted radical prostatectomy: a comprehensive secondary analysis of a public double-blind randomized dataset*. **Journal of Robotic Surgery**. 2026; 20(1):320. DOI: 10.1007/s11701-026-03239-2 | PMID: 41790365 | PMCID: PMC12966217
11. Imam MA, **Hassan AA**, et al. *Coding vs. Culture: How Artificial Intelligence is Transforming Bone, Joint, Skin, and Soft Tissue Infection Management?* **Current Infectious Disease Reports**. 2026; 28(1):2. DOI: 10.1007/s11908-025-00873-z
12. Omran AM, **Hassan AA**, Hamdeno S, Elhendawy MA, Elghor MM, et al. *Sensory Recovery Following Digital Nerve Repair Using Conduits Versus Autologous Nerve Grafts: A Systematic Review and Meta-Analysis*. **Microsurgery**. 2026; 46(1):e70182. DOI: 10.1002/micr.70182 | PMID: 41563884

13. Hassan AA, Elkasaby MH, Megahed HA, Alemam A, Naroz M, et al. *Gastric partitioning compared to conventional gastrojejunostomy as palliative surgeries in patients with gastric outlet obstruction: a pairwise and individual patient data meta-analysis*. **World Journal of Surgical Oncology**. 2026; 24(1):56. DOI: 10.1186/s12957-025-04166-6 | PMID: 41530728 | First author; IPD meta-analysis
14. Elbeny AM, Elkasaby MH, Fouad TA, Soltan ME, Abdelaziz AS, et al.; Hassan AA. *Drug-Coated Balloon Versus Drug-Eluting Stent for Large Coronary Arteries in Patients Presenting With STEMI: A Systematic Review and Meta-Analysis*. **Catheterization and Cardiovascular Interventions**. 2026; 107(1):439–452. DOI: 10.1002/ccd.70358 | PMID: 41277347
15. Hassan AA, Sharaby SA, Elbosraty EE, Abdo MI, Shaban AA, et al. *Comparative Safety and Efficacy of Transcarotid Artery Revascularization (TCAR) Versus Transfemoral Carotid Artery Stenting (TFCAS): A Systematic Review and Meta-Analysis*. **Annals of Vascular Surgery**. 2026; 122:736–754. DOI: 10.1016/j.avsg.2025.08.052 | PMID: 40967268 | First author
16. Abdel Gawad AM, Aboelsaad AY, Elsayed AF, Hassan EMAE, Ashour AY, et al.; Hassan AA. *Standalone 29-MHz micro-ultrasound for classifying clinically significant prostate cancer: a systematic review and diagnostic test accuracy meta-analysis of prospective studies*. **Abdominal Radiology (NY)**. 2026; 51(6):2979–2992. DOI: 10.1007/s00261-025-05218-x | PMID: 41171406 | Senior author; DTA meta-analysis

2025

17. Elmesiry AM, Ibrahim AS, Elabd HA, El Naggar BM, Abd Elsalam EE, et al.; Hassan AA. *Calibrated, explainable machine learning on routine laboratory data to characterize diagnostic assignment patterns in rheumatic diseases: a retrospective study of 12,085 patients*. **BMC Rheumatology**. 2025; 10(1):10. DOI: 10.1186/s41927-025-00607-7 | PMID: 41462380 | Senior author; calibration, SHAP explainability
18. Kandel AM, Elwan AM, Zeen M, Deyab M, Alemam A, et al.; Hassan AA. *Toupet Fundoplication With Re-sleeve for Hiatal Hernia Associated With Failed Sleeve Gastrectomy*. **Cureus**. 2025; 17(11):e96901. DOI: 10.7759/cureus.96901 | PMID: 41409914
19. Hassan AA, Khalafallah MA, Ismail NR, Menshawy E, Mady A, Abouelmagd M, et al. *Intubation conditions and neonatal outcomes with rocuronium versus suxamethonium in cesarean sections: a systematic review and meta-analysis*. **BMC Anesthesiology**. 2025; 25(1):476. DOI: 10.1186/s12871-025-03321-6 | PMID: 41039224 | First author
20. Hassan AA, Soliman AM, Shouman HA, Algammal MI, Salman MF, et al. *Reply to comment on 'Dorsal- vs ventral-onlay buccal mucosal graft urethroplasty: a meta-analysis'*. **BJU International**. 2025 Nov; 136(5):949–950. DOI: 10.1111/bju.16908 | PMID: 40879348
21. Imam M, Hassan A, Peake C, Mitsiou K, Fozo ZA, Antonios T, Khoriaty AA, Narvani A. *The Triple Anatomical Technique for Acromioclavicular Joint Reconstruction: A True Anatomical Reconstruction Technique Using Synthetic Ligaments With Acromioclavicular Ligament Reconstruction*. **Arthroscopy Techniques**. 2025; 14(7):103595. DOI: 10.1016/j.eats.2025.103595 | PMID: 40822197
22. Hassan AA, Mady AM, Abozied H, Algammal MI, Hassan AA, Salman M, et al. *Robotic-assisted vs. open ureteral reimplantation in adults: a systematic review and meta-analysis*. **Journal of Robotic Surgery**. 2025; 19(1):373. DOI: 10.1007/s11701-025-02511-1 | PMID: 40643774 | First author
23. Hassan AA, Soliman AM, Shouman HA, Algammal MI, Salman MF, Hindawy MA, et al. *Dorsal- vs ventral-onlay buccal mucosal graft urethroplasty for urethral strictures: a meta-analysis*. **BJU International**. 2025; 136(3):385–394. DOI: 10.1111/bju.16811 | PMID: 40484913 | First author
24. Hassan AA, et al. *A Machine Learning-Powered Decision Tool to Predict Knee Arthroplasty in Patients with Osteoarthritis*. **International Journal of Artificial Intelligence & Applications (IJAIA)**. 2025. Also presented as an ePoster at ISAKOS Congress 2025, Munich
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CONFERENCE ABSTRACTS, PROCEEDINGS & POSTERS

1. Elkasaby MH, **Hassan AA**, Ahmed MSA. *Safety and Efficacy of Apixaban Compared with Warfarin for Venous Thromboembolism in ESRD: A Meta-Analysis (TH-PO1112)*. **Journal of the American Society of Nephrology**. 2025; 36(10S). ASN Kidney Week 2025. DOI: 10.1681/ASN.2025wq741qsv
2. Elgebaly A, **Hassan AA**, et al. *4P — Machine learning-based prediction of survival in patients with metastatic renal cell carcinoma receiving first-line immunotherapy*. **ESMO Open**. 2025. ESMO Immuno-Oncology Congress 2025. DOI: 10.1016/j.esmoop.2025.104159
3. Mohamed RG, Abouelmagd ME, AbdelMeseh M, **Hassan AA**, et al. *History of Head Trauma Increases the Risk of Multiple Sclerosis Diagnosis: A Meta-analysis of 1,619,640 Participants (P10-1.005)*. **Neurology**. 2025 Apr 8; 104(7 Suppl). AAN Annual Meeting 2025. DOI: 10.1212/WNL.0000000000210688
4. **Hassan AA**, et al. *A Machine Learning-Powered Decision Tool to Predict Knee Arthroplasty in Patients with Osteoarthritis*. **ISAKOS Congress 2025 — ePoster**, Munich, Germany. Presenter ID 184389
5. Gadelmawla AF, et al.; **Hassan AA**. *Abstract 4144899: Invasive vs conservative strategy in management of patients with Non-ST elevation myocardial infarction: a systematic review and meta-analysis of over a million patients*. **Circulation**. 2024; 150(Suppl_1). AHA Scientific Sessions 2024. DOI: 10.1161/CIRC.150.Suppl_1.4144899

6. Abdelaziz A, et al.; **Hassan AA**. Abstract 4146894: Switching to Ticagrelor from clopidogrel after fibrinolytic therapy in patients with ST-elevation myocardial infarction: a meta-analysis of 4334 patients. **Circulation**. 2024; 150(Suppl_1). AHA Scientific Sessions 2024. DOI: 10.1161/CIRC.150.Suppl_1.4146894
7. Abdelaziz A, et al.; **Hassan AA**. Abstract 4146931: The effectiveness of coronary sinus reducer in refractory angina: a meta-analysis of 18 studies. **Circulation**. 2024; 150(Suppl_1). AHA Scientific Sessions 2024. DOI: 10.1161/CIRC.150.Suppl_1.4146931
8. Ismail NR, et al.; **Hassan A**. A network meta-analysis of various treatment modalities for CTEPH. **Atherosclerosis**. 2024 Dec; 399:118666. DOI: 10.1016/j.atherosclerosis.2024.118666
9. Elgebaly A, **Hassan AA**, et al. 44P — Machine learning-based prediction of survival in patients with metastatic renal cell carcinoma receiving first-line immunotherapy. **Immuno-Oncology and Technology**. 2024; 24(Suppl S):101003. DOI: 10.1016/j.iotech.2024.101003

PREPRINTS & PROTOCOL REGISTRATIONS

1. Elgebaly A, **Hassan AA**, et al. Oracle-RCC (Outcome Prediction and Survival Estimation in Renal Cell Carcinoma): development and multi-institutional validation of an artificial-intelligence-based survival prediction model through analysis of 19,579 patients. **SSRN Preprint**, 2025. DOI: 10.2139/ssrn.5103718 | Multi-institutional AI survival model
2. Mohi A, Kareem L, **Hassan AA**. EfficientNet-B0 deep learning model for accurate classification of intrabony lesions in dental panoramic radiographs. **Research Square Preprint**, 2025. DOI: 10.21203/rs.3.rs-6207574/v1 | CNN-based medical imaging classification
3. **Hassan A**, et al. The efficacy and safety of eplerenone compared to spironolactone in patients with heart failure (protocol registration). **OSF Registries**, 2024. DOI: 10.17605/OSF.IO/VNMGK

CERTIFICATIONS & TRAINING

Clinical & Emergency Care

- Basic Life Support (BLS)
- Advanced Cardiovascular Life Support (ACLS)

Clinical Research & Governance

- Good Clinical Practice (GCP) / Clinical Research Training
- Negida Academy — Research Certification
- Clinical Workshop Certification — Histology Department, Al-Azhar Medical School
- Abdominal ultrasound course (certificate on file)

Data Science, AI & Software

- IBM Data Science Professional Certificate
- IBM Data Analyst Professional Certificate
- IBM AI Engineering Professional Certificate
- IBM AI Developer Professional Certificate
- Meta Back-End Developer Professional Certificate
- Building Deep Learning Applications with Keras 2.0
- R for Data Science: Analysis & Visualization

LANGUAGES

- Arabic — Native
- English — Professional working proficiency

- German — Basic

MEMBERSHIPS, COLLABORATORS & NETWORKS

- **ISAKOS Medical Student Member** (International Society of Arthroscopy, Knee Surgery, and Orthopaedic Sports Medicine) — presenter ID 184389.
- **ResBot.ai** — Prof. Mohamed A. Imam (University of East London), Dr. Ahmed Negida (Harvard Medical School).
- **OrthoGlobe Initiative (London)** — Prof. Mohamed A. Imam, Tony Antonios, Al-Achraf Khoriaty, Ali Narvani.
- **Medical Research Group of Egypt (MRGE) / Negida Academy** — cross-institutional SR/MA collaborations.
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CAREER OBJECTIVE

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SUPPORTING DOCUMENTS AVAILABLE ON REQUEST

- Passport bio page
- Egyptian national ID (front and back)
- Birth certificate
- MBChB degree certificate (Al-Azhar University, serial HH0171494, issued 2 July 2024)
- Ministry of Higher Education attestation of the degree
- Internship completion certificate no. 288 (Al-Azhar University Hospitals, 28 February 2025)
- Egyptian Ministry of Health practice license card (front and back)
- Egyptian Medical Syndicate registration
- Dataflow Primary Source Verification report (case S003-2504-2401814, POSITIVE)
- Cumulative transcript / grade sheet (Al-Azhar Faculty of Medicine)
- Passport-size photograph
- R for Data Science course certificate (PDF)